Exercise 14-1.
請寫一個 Python 程式 `week14_1.py`
題目要求:

1. 建立父類別 **Vehicle:**

- 屬性:`brand`(品牌名稱)
- 方法:`move()` → 回傳 `"Moving..."`

2. 建立兩個子類別:

- **Car**:覆寫 `move()` → 回傳 `"Car is driving"`
- **Bike**:覆寫 `move()` → 回傳 `"Bike is pedaling"`

(範例輸入輸出參考下圖, 綠色表示系統輸出, 黑色表示程式碼參考)

```
None
vehicles = [
    Car("Toyota"),
    Bike("Giant"),
    Car("Tesla")
]

Toyota: Car is driving
Giant: Bike is pedaling
Tesla: Car is driving
```

```python
class vehical:  2 usages
    def __init__(self, brand):
        self.brand = brand
    def move(self):
        return "Moving..."

class car(vehical):  2 usages
    def move(self):  1 usage
        return "Car is driving"

class bike(vehical):  1 usage
    def move(self):  1 usage
        return "Bike is driving"

vehical = [
    car("Toyota"),
    bike("Giant"),
    car("Tesla")
]

for a in vehical:
    print(f"{a.brand} : {a.move()}")
```

```
C:\Users\benss\PycharmProjects\PythonProj
Toyota : Car is driving
Giant : Bike is driving
Tesla : Car is driving
```

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Exercise 14-2.
請寫一個 Python 程式 `week14_2.py`
題目要求:
設計一個 圖書館借書系統 **Library System**:
LibrarySystem 能接收一個整數 `capacity`, 代表圖書館最多能容納多少本書。
初始化時, 會傳入一個整數, 代表目前圖書館能容納書本數量的上限。
LibrarySystem 的方法:

1. **addBook(title):**
   若圖書館仍有空位, 加入書籍, 回傳 **True;** 若滿了, 回傳 **False**。

2. **borrowBook(title):**
   若書存在 → 借出並從館內移除, 回傳 **True;** 若書不存在 → 回傳 **False**。

3. **returnBook(title):**
   若仍有空位 → 加回書籍, 回傳 **True;** 若沒有空位 → 回傳 **False**。

```
None
===input===
["LibrarySystem", "addBook", "addBook", "borrowBook", "returnBook",
"addBook"]
[[2], ["Python"], ["Math"], ["Python"], ["Python"], ["History"]]

===output===
[None, True, True, True, True, False]
```

```python
import sys
import json
import ast


class LibrarySystem:  1 usage
    def __init__(self, capacity: int):
        self.capacity = capacity
        self.books = []  # 目前館藏(可借)書名清單，可允許重複書名

    def addBook(self, title: str) -> bool:  1 usage
        # 若仍有空位 -> 加入書籍，回傳 True；若滿了 -> 回傳 False
        if len(self.books) < self.capacity:
            self.books.append(title)
            return True
        return False

    def borrowBook(self, title: str) -> bool:  1 usage
        # 若書存在 -> 借出並從館藏移除，回傳 True；不存在 -> 回傳 False
        try:
            self.books.remove(title)  # 移除第一本符合的書
            return True
        except ValueError:
            return False
```

```python
    def returnBook(self, title: str) -> bool:  1 usage
        # 若仍有空位 -> 歸還(加回)書籍，回傳 True；若沒有空位 -> 回傳 False
        if len(self.books) < self.capacity:
            self.books.append(title)
            return True
        return False


def _extract_two_arrays(text: str):  1 usage
    """從整段輸入中抓出前兩個 [ ... ] 區塊，支援中間夾雜 ===input=== 之類文字。"""
    arrays = []
    i = 0
    n = len(text)

    while i < n and len(arrays) < 2:
        if text[i] == "[":
            depth = 0
            j = i
            while j < n:
                if text[j] == "[":
                    depth += 1
                elif text[j] == "]":
                    depth -= 1
                    if depth == 0:
```

```python
                        arrays.append(text[i : j + 1].strip())
                        i = j
                        break
                j += 1
        i += 1

    if len(arrays) < 2:
        raise ValueError("Input must contain two array blocks like: [..]\\n[..]")

    def load(s: str):
        try:
            return json.loads(s)        # JSON 風格
        except Exception:
            return ast.literal_eval(s)  # Python list 風格

    return load(arrays[0]), load(arrays[1])


def main():  1 usage
    data = sys.stdin.read()
    if not data.strip():
        return

    ops, args = _extract_two_arrays(data)
```

```python
    lib = None
    output = []

    for k in range(len(ops)):
        op = ops[k]
        arg = args[k]

        if op == "LibrarySystem":
            lib = LibrarySystem(arg[0])
            output.append(None)
        elif op == "addBook":
            output.append(lib.addBook(arg[0]))
        elif op == "borrowBook":
            output.append(lib.borrowBook(arg[0]))
        elif op == "returnBook":
            output.append(lib.returnBook(arg[0]))
        else:
            raise ValueError(f"Unknown operation: {op}")

    print(output)


if __name__ == "__main__":
    main()
```

```
C:\Users\benss\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users
["LibrarySystem", "addBook", "addBook", "borrowBook", "returnBook",
"addBook"]
[[2], ["Python"], ["Math"], ["Python"], ["Python"], ["History"]]
===output===
[None, True, True, True, True, False]
```

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